

It's Morphin' Time!

Yield to the Computer Chip that can be moulded into anything you want

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And a dark time came, for RAM was scarce, and the swap was reduced to naught. And the OS said unto the Chip, "Be Memory!", and lo! For the Chip was now Memory.

— *The E-Testament, Chaos 21:12*

All right, so there's no such thing as the E-Testament. But imagine a time when you are running the immensely taxing *Doom V*, and your system finds itself desperately out of memory. All it needs to do now is tell one of the other chips to transform itself into a memory chip, and all is well in the gaming world again. Even as these words spill out, the idea seems quite absurd. After all, whoever heard of hardware running around transforming itself into new components? We ranted about adaptive software not too long ago (*The Lizard of Oz*, Digit, January 2006); how about hardware that can adapt itself to the demands of the situation? If William Ditto, chair of the Department of Biomedical Engineering at the University of Florida, and his colleagues, have their heads screwed on right (and many, including the US Navy, believe they do), we might actually see a prototype chip that can transform itself from CPU to memory to graphics card to take-your-pick as early as next year! The idea is called Chaos Computing, and will, as the name suggests, use a phenomenon that scientists are otherwise known to hate—chaos.

The Butterfly Effect And Other Stories

Chaos, in the scientific sense, isn't quite the same as we know it in real life. While life's chaos is completely random and unpredict-

able, science defines a chaotic system to be one that is quite predictable, but extremely sensitive to its initial conditions. To understand the concept better, try this: get your hands on a crazy ball (those little unnaturally bouncy balls) and bounce it down a sufficiently bumpy slope. If you can find it after that, repeat the exercise *exactly* as you did it before, right down to the movement of your arm. Most likely, it won't follow the same path down as it did before; in fact, it will probably go so astonishingly off course, you'd think it has a mind of its own. This is because even the slightest extra movement or the tiniest irregularity on the surface of the ball has a dramatic effect on the way it comes up from the first bounce. This then affects the next bounce and the one after that, and so on, and now nobody will believe your "exact reproduction" story. Note, however, that calculating these paths is quite possible, though horribly complex, so the system isn't really random—just fickle.

Such fickleness is what led Edward Lorenz, American mathematician and meteorologist, to talk about the *Butterfly Effect* while studying chaos in weather. It's a little like the Domino Effect, only with a bigger, meaner domino falling each time. The tiniest change in his simulation caused major changes in the weather system, and his subsequent paper read, "One meteorologist remarked that if the theory were correct, one flap of a seagull's wings could change the course of weather forever." The seagull was soon replaced by the prettier butterfly.

Chaos And The Chip

The main idea behind Chaos Computing is the manipulation of *Logic Gates*, the basic elements of computing. You might have heard of these—the AND gate, for example, outputs a binary 1 only if all its inputs (A and B and C and so on)

